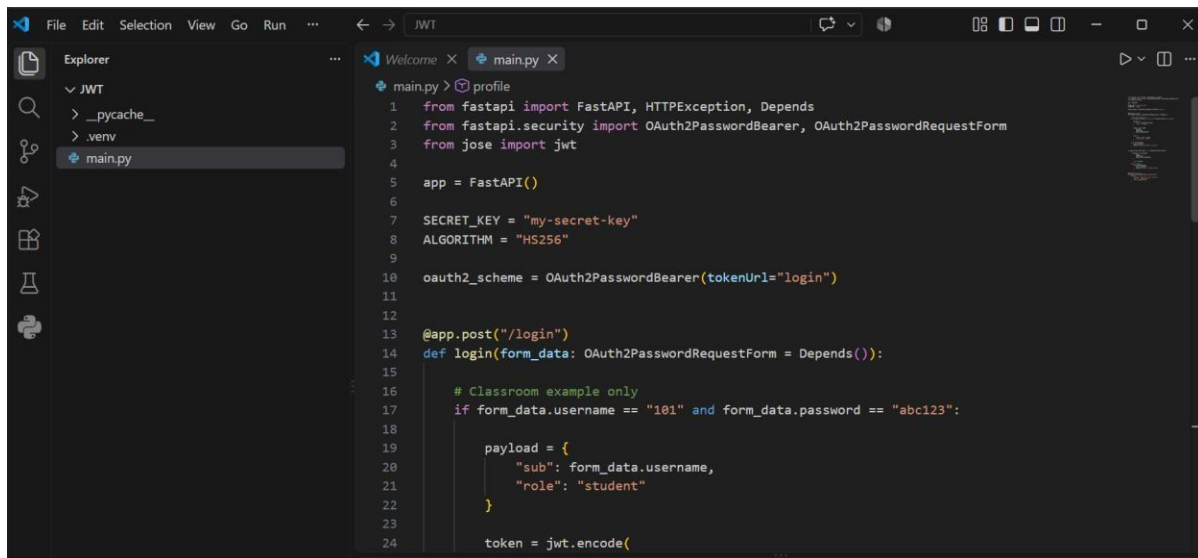
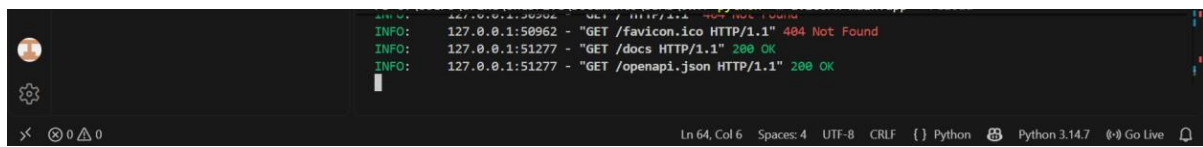


SKILL-6 HARSHITHA.B



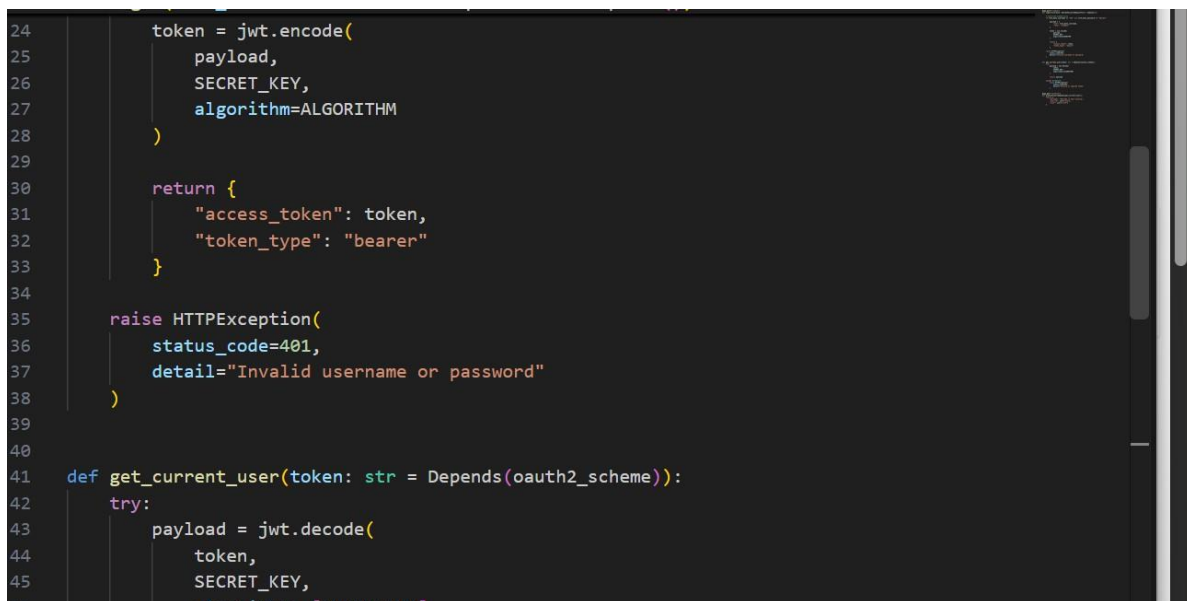
The screenshot shows the Visual Studio Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with a folder named 'JWT' containing files like 'main.py', 'profile', and '._pycache_'. The code editor displays the 'main.py' file, which is a FastAPI application. The code includes imports for FastAPI, OAuth2PasswordBearer, OAuth2PasswordRequestForm, and jwt. It defines an app, sets a secret key and algorithm, and creates an OAuth2PasswordBearer instance. A login endpoint is defined with a POST method, and a login function is implemented that checks the username and password. If the credentials are correct, it generates a JWT token and returns it. The code is as follows:

```
1 from fastapi import FastAPI, HTTPException, Depends
2 from fastapi.security import OAuth2PasswordBearer, OAuth2PasswordRequestForm
3 from jose import jwt
4
5 app = FastAPI()
6
7 SECRET_KEY = "my-secret-key"
8 ALGORITHM = "HS256"
9
10 oauth2_scheme = OAuth2PasswordBearer(tokenUrl="login")
11
12
13 @app.post("/login")
14 def login(form_data: OAuth2PasswordRequestForm = Depends()):
15
16     # Classroom example only
17     if form_data.username == "101" and form_data.password == "abc123":
18
19         payload = {
20             "sub": form_data.username,
21             "role": "student"
22         }
23
24         token = jwt.encode(
```



The screenshot shows the terminal output of the application. It displays three log messages: a 404 Not Found response for a GET request to /favicon.ico, and two 200 OK responses for GET requests to /docs and /openapi.json. The status bar at the bottom indicates the file is at line 64, column 6, with 4 spaces, UTF-8 encoding, CRLF line endings, and Python 3.14.7 interpreter.

```
INFO: 127.0.0.1:50962 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO: 127.0.0.1:51277 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:51277 - "GET /openapi.json HTTP/1.1" 200 OK
```



The screenshot shows the continuation of the 'main.py' file in the Visual Studio Code editor. The code continues from the previous block, showing the completion of the login function and the start of the 'get_current_user' function. The login function raises an HTTPException with a status code of 401 and a detail message 'Invalid username or password' if the credentials are incorrect. The 'get_current_user' function is defined to take a token and return the user information. The code is as follows:

```
24         token = jwt.encode(
25             payload,
26             SECRET_KEY,
27             algorithm=ALGORITHM
28         )
29
30         return {
31             "access_token": token,
32             "token_type": "bearer"
33         }
34
35     raise HTTPException(
36         status_code=401,
37         detail="Invalid username or password"
38     )
39
40
41 def get_current_user(token: str = Depends(oauth2_scheme)):
42     try:
43         payload = jwt.decode(
44             token,
45             SECRET_KEY,
```

```

main.py > profile
41 def get_current_user(token: str = Depends(oauth2_scheme)):
42     try:
43         payload = jwt.decode(
44             token,
45             SECRET_KEY,
46             algorithms=[ALGORITHM]
47         )
48
49         return payload
50
51     except Exception:
52         raise HTTPException(
53             status_code=401,
54             detail="Invalid or expired token"
55         )
56
57

```

```

57
58 @app.get("/profile")
59 def profile(user=Depends(get_current_user)):
60     return {
61         "message": "Welcome to your profile",
62         "user_id": user["sub"],
63         "role": user["role"]
64     }

```

POST OR LOGIN

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/login' \
  -H 'accept: application/json' \
  -H 'Content-type: application/x-www-form-urlencoded' \
  -d 'grant_type=password&username=101&password=abc123&scope=&client_id=string&client_secret=*****'
```

Request URL

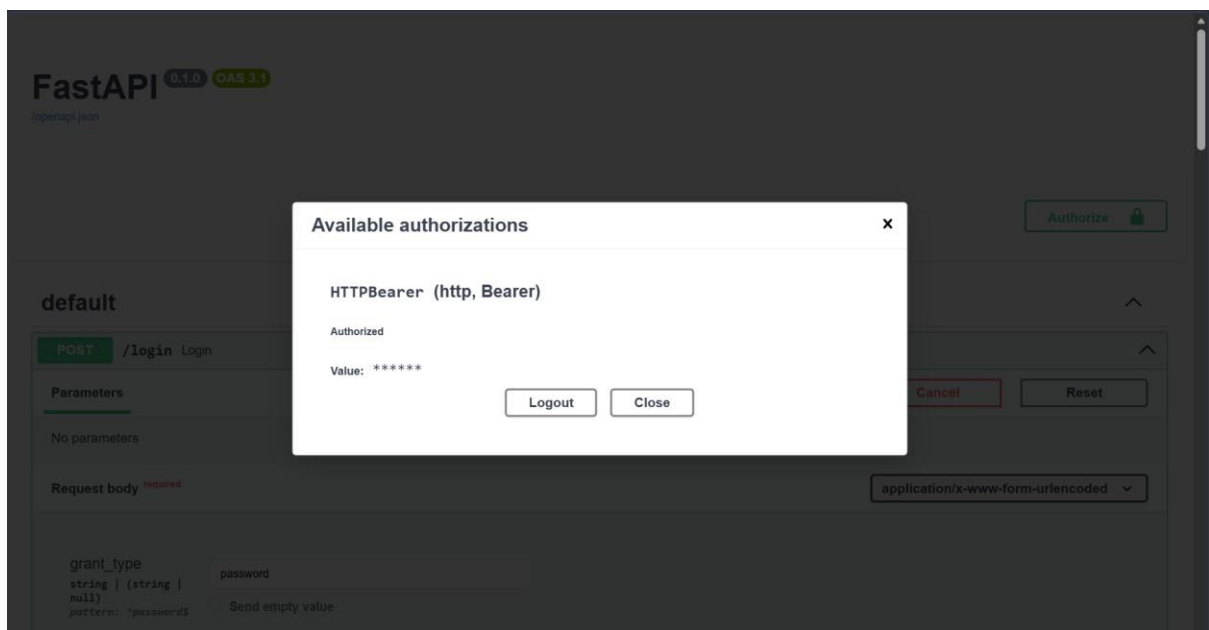
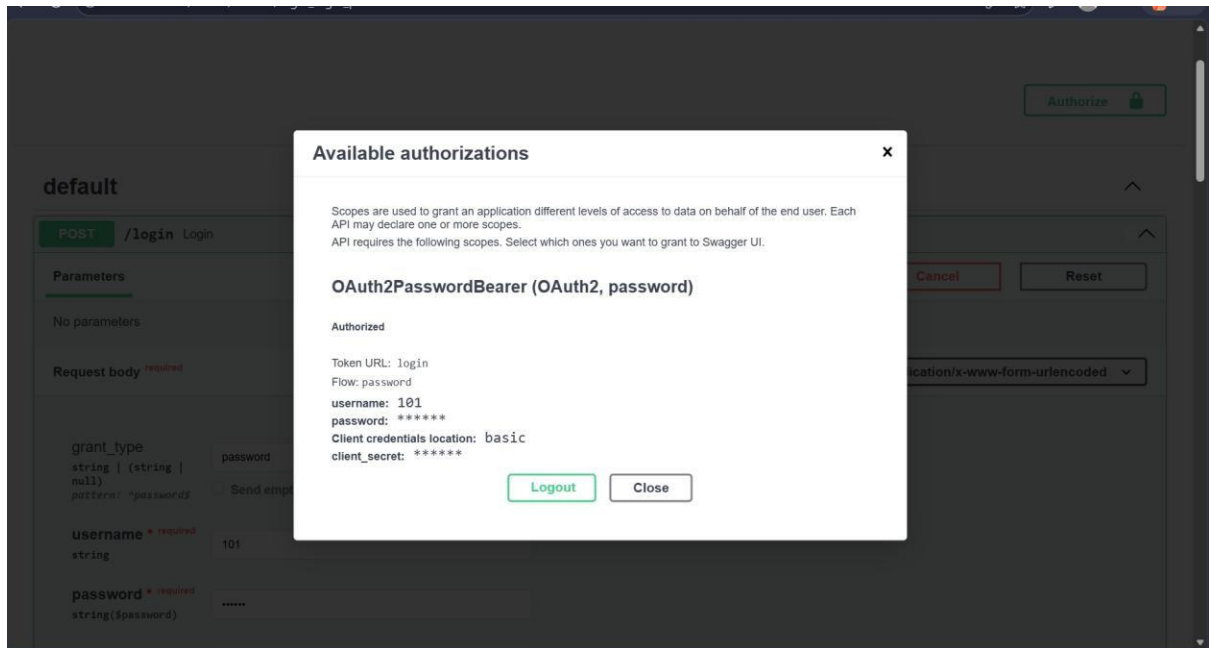
```
http://127.0.0.1:8000/login
```

Server response

Code	Details
200	Response body <pre>{ "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMDE1LCJyb2kiOiJ1b3R1ZGVudCJ9.NHqwkDLP8iQ5BFQ1iKmdowVLOjBLUMHm4aRQ6MK1zI", "token_type": "bearer" }</pre> Download
	Response headers <pre>content-length: 162 content-type: application/json date: Wed, 16 Sep 2026 09:48:52 GMT server: uvicorn</pre>

Responses

Code	Description	Links
------	-------------	-------



GET

/profile Profile

Cancel

Parameters

No parameters

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/profile' \
  -H 'accept: application/json' \
  -H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMDE1LCJyb2kiOiJoIjoic3R1ZGVudCJ9.NHqwkDLPB1Q58FQ11KmdowvLoJbLUMdHm4aRQ6MK1zI'
```

Request URL

```
http://127.0.0.1:8000/profile
```

Server response

Code	Details
200	Response body<pre>{ "message": "Welcome to your profile", "user_id": "101", "role": "student" }</pre> Download